GLA University, Mathura

End- Term Examination, 2011-12

Course: - B.Tech. II-Year, III-Sem.

Subject: - Mathematics **Uni. Roll No:-**

Time:-2HOURS 30 Minutes **Total Marks:-80**

Notes:-

1) Attempt **ALL** questions of Section- A, **ANY FOUR** from Section- B ,and **ANY FOUR** from Section- C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

Section:-A $(10 \times 2 = 20)$

Q .1 Cauchy – Riemann equations in polar coordinates .....

Q.2 Value of integral $\int_c \frac{e^{-z}}{z+1} dz$ Where C is the circle $|z| = 1/2$ .....

Q.3 If $f(z)$ have a singular point at $z = a$ then it can be expanded by Taylor's Series or by Laurent's Series .....

Q.4 $f(z) = \sin\frac{1}{1-z}$ at $z = 1$ have the singularity of the type.....

Q.5 Residue of $f(z) = \frac{z^3}{z^2-1}$ at $z=\infty$ is.....

Q.6 Order of each poles of the function $f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$ .....

Q.7 Write the formula for fourier sine transform.....

Q.8 If $f(p)$ is the complex Fourier transform of $F(x)$ then $F[F(ax)] =$.....

Q.9 Z-Transform of unit impulse function is .....

Q-10 If $Z\{f(k)\} = F(z)$, Then $Z\{k^n f(k)\}$ is .....

Section:-B $(4 \times 5 = 20)$

Q1. Prove that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic.

Q.2 Solve the biquadratic equation $x^4 - 3x^2 - 42x - 40 = 0$ by Ferrari's method.

Q.3 Find the Fourier transform of $F(x) = e^{-|x|}$

Q.4 Solve by Z-transform $y_{k+1} + y_k = 1$ if $y_0 = 0$

Q.5 Find the Fourier series of $f(x) = x^2$, $-\pi \leq x \leq \pi$.

Secion.- C $(4 \times 10 = 40)$

Q1. Prove that the function F(z) defined by

$$F(z) = [x^3(1+i) - y^3(1-i)] / [x^2 + y^2] \quad ; \; z \neq 0 \text{ and } F(0) = 0$$

is not regular at the origin although Cauchy – Riemann equations are satisfied there.

Q .2 If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and

$u - v = (x - y)(x^2 + 4xy + y^2)$ then construct f(z) in terms of z .

Q .3 Determine the poles of the function $f(z) = \frac{z-1}{(z+1)^2(z-2)}$ and hence evaluate

$\int_C f(z)\,dz$ by Cauchy's residue theorem where C is the circle $|z - i| = 2$

Q .4 Obtain the Taylor's series expansion of $f(z) = \frac{1}{z^2 - 4z + 3}$ about the point $z = 4$.

Find its region of convergence.

Q .5 Solve the integral $\int_0^{2\pi} \frac{d\theta}{5 - 3\cos\theta}$ using contour integration.

GLA University, Mathura

Final Term Examination, 2011-12

Course: - B.Tech. II-Year, III-Sem.

Subject: - Data Structure And Algorithm **Uni. Roll No:-**

Time:-2 Hrs 30 Min. **Total Marks:-80**

Notes:-

1) Answer all questions, Group A is **compulsory, Any Four** from Group B and **Any three** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

GROUP/SECTION-A(1x20)

Select the correct answer:-

Q.1 Let A be an adjacency matrix of a graph G. The th ij entry in the matrix K A , gives
(A) The number of paths of length K from vertex Vi to vertex Vj.
(B) Shortest path of K edges from vertex Vi to vertex Vj.
(C) Length of a Eulerian path from vertex Vi to vertex Vj.
(D) Length of a Hamiltonian cycle from vertex Vi to vertex Vj.

Q.2 The OS of a computer may periodically collect all the free memory space to form contiguous block of free space. This is called
(A) Concatenation **(B)** Garbage collection
(C) Collision **(D)** Dynamic Memory Allocation

Q.3 You have to sort a list L consisting of a sorted list followed by a few "random" elements.Which of the following sorting methods would be especially suitable for such a task?
(A) Bubble sort **(B)** Selection sort
(C) Quick sort **(D)** Insertion sort

Q.4 B Trees are generally
(A) very deep and narrow **(B)** very wide and shallow
(C) very deep and very wide **(D)** cannot say

Q.5 A technique for direct search is
(A) Binary Search **(B)** Linear Search
(C) Tree Search **(D)** Hashing

Q.6 If a node having two children is deleted from a binary tree, it is replaced by its
(A) Inorder predecessor **(B)** Inorder successor
(C) Preorder predecessor **(D)** None of the above

Q.7 The number of interchanges required to sort 5, 1, 6, 2 4 in ascending order using Bubble Sort is
(A) 6 **(B)** 5
(C) 7 **(D)** 8

Q.8 The postfix form of the expression (A+ B)*(C*D– E)*F / G is

(A) AB+ CD*E – FG /** **(B)** AB + CD* E – F **G /

(C) AB + CD* E – *F *G / **(D)** AB + CDE * – * F *G /

Q.9 The complexity of multiplying two matrices of order m*n and n*p is

(A) mnp **(B)** mp **(C)** mn **(D)** np

Q.10 In worst case Quick Sort has order

(A) O (n log n) **(B)** O (n2/2) **(C)** O (log n) **(D)** O (n2/4)

Q.11 A full binary tree with 2n+1 nodes contain

(A) n leaf nodes **(B)** n non-leaf nodes

(C) n-1 leaf nodes **(D)** n-1 non-leaf nodes

Q.12 If a node in a BST has two children, then its inorder predecessor has

(A) no left child **(B)** no right child **(C)** two children **(D)** no child

Q.13 A binary tree in which if all its levels except possibly the last, have the maximum number of nodes and all the nodes at the last level appear as far left as possible, is known as

(A) full binary tree. **(B)** AVL tree. **(C)** threaded tree. **(D)** complete binary tree.

Q.14 A linear list of elements in which deletion can be done from one end (front) and insertion can take place only at the other end (rear) is known as a

(A) queue. **(B)** stack. **(C)** tree. **(D)** linked list.

Q.15 An undirected graph G with n vertices and e edges is represented by adjacency list. What is the time required to generate all the connected components?

(A) O (n) **(B)** O (e) **(C)** O (e+n) **(D)** O (2) e

Q.16 Consider a linked list of n elements. What is the time taken to insert an element after an element pointed by some pointer?

(A) O (1) **(B)** O (log2 n) **(C)** O (n) **(D)** O (n log2 n)

Q.17 The smallest element of an array's index is called its

(A) lower bound. **(B)** upper bound. **(C)** range. **(D)** extraction.

Q.18 In a circular linked list

(A) components are all linked together in some sequential manner.

(B) there is no beginning and no end.

(C) components are arranged hierarchically.

(D) forward and backward traversal within the list is permitted.

Q.19 One of the major drawback of B-Tree is the difficulty of traversing the keys sequentially.

(A) True **(B)** False

Q.20 O(N) (linear time) is better than O(1) constant time.

(A) True **(B)** False

Section B

Q1. a) A two dimensional array TABLE [6] [8] is stored in row major order with base address 351. What is the address of TABLE [3] [4]? (2)

b) What is an algorithm? What are the characteristics of a good algorithm? (3)

Q2: What are expression trees? Represent the following expression using a tree. Comment on the result that you get when this tree is traversed in Preorder, Inorder and postorder. (a-b) / ((c*d)+e) (5)

Q3: Show the result of inserting the keys. (5)

F, S, Q, K, C, L, H, T, V, W, M, R, N , P, A, B, X, Y, D, Z, E in the order to an empty B-tree of degree-3.

Q4: Which are the two standard ways of traversing a graph? Explain them with an example of each. (5)

Q5: Write an algorithm to create a single linked list. Write functions to do the following Operations (1+2+2)

(i) Insert a new node at the end

(ii) Delete the first node

Section C

Note:Question number 1is compulsory

Q1:a)The following values are to be stored in a hash table (7+7)

25, 42, 96, 101, 102, 162, 197

Describe how the values are hashed by using division method of hashing with a table size of 7. Use chaining as the method of collision resolution.

b) What do you mean by hashing? Explain any three popular hash functions.

Q2:a)What do you understand by file organization? List the various file operations. Explain them. (7+6)

b) Write binary search algorithm and trace to search element 91 in following list:
13 30 62 73 81 88 91

What are the limitations of Binary Search?

Q3:a) Write down the algorithm of quick sort. Also write its complexity for all the cases.

b) Sort the following array of elements using quick sort (show pass-wise). (7+6)

{3 1 4 1 5 9 2 6 5 3 5 8}

Q4: a) Write down the algorithm of Merge sort. Also write its complexity for all the cases. (6+7)

b) Sort the following sequence of keys using merge sort(show pass-wise).
66, 77, 11, 88, 99, 22, 33, 44, 55

GLA UNIVERSITY MATHURA
END TERM EXAMINATION, 2011-12
COURSE: - B.TECH II-YEAR, III-SEM

SUBJECT: DISCRETE MATHEMATICS (CSE213) **UNI. ROLL NO:**

TIME: 2HRS 30 MIN **MAX MARKS: 80**

Note:-

1. Answer **all questions** from Group A, **Any Four** from Group B and **Any Four** from Group C.
2. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

GROUP A

Q1. Answer all the questions of this group. **[1 x 20 = 20]**

I. The probability that a number selected at random between 100 and 999 (both inclusive) will not contain the digit 7 is .....

II. A class is composed of 2 brothers and 6 other guys. The number of ways all the boys can be seated in a round table so that two brothers are not seated together is .....

III. Given the relation R = {(1,2),(2,3)}.The minimum number of ordered pairs that must be added to this set so that the enlarged relation becomes reflexive, symmetric and transitive is .....

IV. Consider the mapping f:X $\rightarrow$ Y. f is a bijection if and only if
(a) f(x) = f(y) implies x = y for all x, y (b) range of f is Y
(c) both (a) and (b) are true (d) the co-domain equals to range

V. Let A and B be two arbitrary events. Then
(a) $P(A \cap B) = P(A) + P(B)$ (b) $P(A \cup B) = P(A)+P(B)$
(c) $P(A/B) = P(A \cap B) + P(B)$ (d) $P(A \cup B) \leq P(A)+P(B)$

VI. The subset of a countable set
(a) has to be countable (b) may or may not be countable
(c) has to be finite (d) none of the above

VII. Every element of some ring(R,+,*),is such that a *a = a. The ring
(a) is commutative (b) is not commutative
(c) may or may not be commutative (d) none of the above

VIII. The number of subsets of {1,2...n} of odd cardinality is
(a) Dependant on the value of n (b) 2^{n-1} if n is odd
(c) 2^{n-1} if n is even (d) 2^{n-1} for any value of n

IX. The set {1, 2, 3, 4, 6, 8, 12, 24} together with GCD as binary operation is not a group because
(a) it is not associative (b) identity does not exist
(c) inverse is not unique (d) inverse does not exist

X. If the proposition $\neg P$ is true, then the truth value of the proposition $\neg P \vee (P \Rightarrow Q)$ is
(a) true (b) multi-valued (c) false (d) cannot be determined

XI. Translate the following statement into symbols using quantifiers, variables and predicate symbols
" All men are giants."

XII. Translate the following statement into symbols using quantifiers, variables and predicate symbols
"There is a student who likes mathematics but not history."

XIII. Determine the truth or falsity of the following sentence where universe is the set of integers
$\exists x, [\{x^2 > 10\} \wedge x \text{ is even}]$

XIV. Determine the truth or falsity of the following sentence where universe is the set of integers:
$\forall x, \{\exists y, [x^2 = y]\}$

XV. The total number of possible Boolean functions involving n Boolean variables is
(a) infinitely many (b) n^n (c) n^2 (d) none of these

XVI. The XOR operator is
(a) commutative (b) Associative (c) distributive over AND operator (d) none of these

XVII. $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow \wedge S)$ is equivalent to
(a) $S \wedge R$ (b) $S \rightarrow R$ (c) $S \vee R$ (d) none of these

XVIII. Which of the following is a tautology?
(a) $((P \vee Q) \wedge Q) \Leftrightarrow Q$ (b) $(P \vee (P \rightarrow Q)) \rightarrow P$ (c) $((P \vee Q) \wedge P) \rightarrow Q$ (d) $((P \vee Q) \wedge \neg P)$

XIX. Identify the valid conclusion from the premises $P \vee Q$, $Q \rightarrow R$, $P \rightarrow M$, $\neg M$
(a) $P \wedge (Q \vee R)$ (b) $P \wedge (Q \wedge R)$ (c) $R \wedge (P \vee Q)$ (d) $Q \wedge (P \vee R) \wedge M$

XX. In the set of integers, a relation R is defined as a R b if and only if $b = |a|$. This relation is
(a) reflexive (b) irreflexive (c) symmetric (d) anti- symmetric

GROUP B

Q2. Answer any FOUR questions of this group. **[5 x 4 = 20]**

I. A relation R is defined on the set of integers Z in the following way:
$(x, y) \varepsilon R \Leftrightarrow x^2 + y^2 = 25$
Find R and R^{-1} as the sets of ordered pairs and hence find their respective domains.

II. Show that if a and b are any two elements of a group G, then $(ab)^2 = a^2b^2$ if and only if G is abelian.

III. In a survey of 600 television viewers, the following information was obtained: 385 watch cricket matches, 295 watch hockey matches, 215 watch football matches, 145 watch cricket and football matches, 170 watch cricket and hockey matches, 150 watch hockey and football matches and 150 do not watch any of the three kinds of matches.
(a) How many people in survey watch all three kinds of matches?
(b) How many people watch exactly one of the sports?

IV. A candidate is required to answer 6 out of 10 questions which are divided into two groups each containing five questions and he is not permitted to attempt more than four from any group. In how many ways can he make up his choice?

V. Three machines M1, M2 and M3 produce identical items. Of their respective outputs 5%, 4% and 3% of the items are defective. On a certain day, M1 has produced 25% of the total output, M2 has produced 30% and M3 the remainder. An item selected at random is found to be faulty. What are the chances that it was produced by the machine with highest output?

GROUP C

Q3. Answer any FOUR questions of this group. **[4 x 10 = 40]**

I. If R is a ring such that $a^2 = a \; \forall \; a \varepsilon R$, then prove that
(a) $a + a = 0 \; \forall a \varepsilon R$ i.e. each element of R is its own additive inverse.
(b) $a + b = 0 \Rightarrow a = b$
(c) R is a commutative ring.

II. State and prove Lagrange's theorem.

III. (a) Define and draw the following graphs:
(i) a regular graph (ii) a complete graph
(iii) a bipartite graph (iv) complement of the cycle graph c_4
(b) Prove that if a graph, which may be connected or disconnected, has exactly two vertices of odd degree, then there must be a path joining these two vertices.

IV. Prove that a group G is abelian if and only if
$(ab)^{-1} = a^1 b^{-1} \; \forall \; a,b \varepsilon G$

V. Show by means of examples that the union of two subgroups may or may not be a subgroup.

GLA University, Mathura
End Sem Examination, 2011-12
Course: - B.Tech (CS) II-Year, III-Sem.

Subject: -Computer Organization **Uni. Roll No:-**
Time:-2hr 30min **Total Marks:-80**
Notes:-

1) Answer all questions, Group A is **compulsory, Any four** from Group B and **Any four** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
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5) Figures on the right-hand side margin indicate full marks.

GROUP/SECTION-A

I Fill in the blanks **(20)**

1) A Combinational circuit that forms the arithmetic sum of the three input bits is called
   a) Full adder b) Half adder c) Flip flop d) Multiplexer

2) Memory unit that communicates directly with the CPU is called
   a) Auxiliary memory b) Main memory c) cache memory d) multiprogramming.

3) The task of performing arithmetic and logical operations is called
   a) editing b) storage c) output d) ALU

4) The process of producing results from the data for getting useful information
   a) Output b) Input c) Processing d) Storage

5) Interrupts which are initiated by an instruction are
   a) Internal b) External c) Hardware d) Software

6) CPU does not perform operation
   a) Data transfer b) logic operation c)arithmetic operation d) all of these

7) A collection of 8 bits is called
   a) Byte b) Bit c) word d) record

8) A micro program is a sequencer perform the operation
   a) Read b) write c) read and execute d) read and write

9) How many address lines are needed to address each memory locations in a 2048 x 4 memory chip?
   a) 10 b) 11 c) 8 d) 12

10) Which of the following registers is used to keep track of address of the memory location where the next instruction is located?
   a) Memory Address Register b) Memory Data Register
   c) Instruction Register d) Program Register.

II True or false

1) RAM is called DRAM(Dynamic RAM) when it can do several things simultaneously.
2) The two's complement representation of –10 is 11110110.

3) Floating point representation is used to store a real number.
4) Cache memory refers to fast memory present on the processor chip that is used to store recently accessed data.
5) Binary numbers can be used to represent both fractions and integers.

III Fill in the blanks

1) Input or output device attached to the computer are also called .....
2) An ..... Is a digital circuit that performs the inverse operation of a decoder.
3) A register capable of shifting its binary information in one or both directions is called a .....
4) A ..... is a elementary operation performed on the information stored in one or more registers.
5) The control variable at any given time can be represented by a string of 1's and 0's called a.....

GROUP/SECTION-B

II Attempt any four **(5*4)** **(20)**

Q1) Simplify the Boolean function in sum-of-product form by means of four variable map. Draw the logic diagram with NAND gate.
$F(A,B,C,D) = \sum(0, 2, 8, 9, 10, 11, 14, 15)$

Q2) Explain the different cycles of an instruction execution.

Q3) Define the following :
a) Micro operation b) Microinstruction c) Micro program d) Micro code

Q4) Explain decoders. Draw the block diagram of 2 to 4 line decoder with NAND gate. Also show its truth table.

Q5) Implement the following function
$F = x'y'z' + xy' + z$ using NOR gates only.

GROUP/SECTION-C

III Attempt any four **(10*4)** **(40)**

Q1) What do you mean by modes of transfer? Explain the following:
a) Programmed I/O b)Interrupt Initiated I/O

Q2) Describe DMA with suitable block diagram. Why does DMA have priority over the CPU when both request a memory transfer? Explain

Q3) What is Cache memory? Explain it with its basic characteristics .How Associative mapping and direct mapping is considered in the organization of cache memory.

Q4) Write a short notes on any three of the following
a) Auxiliary memory b)Memory Hierarchy c)Virtual memory d) Magnetic disk

Q5) Describe asynchronous data transfer. What are the methods through which it can be achieved? Explain stroke control and handshaking.

GLA University, Mathura

END Term Examination, 2011-12

Course: - B.Tech **Year:II** **Sem:III**

Subject: Object Oriented Programming

Uni. Roll No:-

Duration:-2:30 Hours **Total Marks:-80**

Notes:-

1) Answer all questions, Group A is **compulsory, Any Four** from Group B & **Any Four** from Group C. Each Question of Group A carry 1 Marks
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

Section A

Objective Type Questions: **(20*1=20)**

1. Object based Programming languages does not include the concept of
   (a) Encapsulation (b) Data Hiding (c) Dynamic binding (d) All of these
2. Object Modelling describes the _____ structure of a system
   (a)Dynamic (b) Static (c) Computational (d) Both (b) and (c)
3. C++ was developed in the year
   (a)1972 (b) 1975 (c)1980 (d)1988
4. Which of the following is not a C++ keyword?
   (a) class (b) delete (c)operator (d)type
5. How many bytes are occupied by the string literal constant "XYZ" in the memory
   (a) 1 (b)2 (c)3 (d)4
6. We provide a default value to arguments (in concept of default parameters) from
   (a) Left to Right (b) Right to Left (c) Middle (d) From any where
7. Static member Functions are called by using _____ operator
   (a) Dot (b) Scope Resolution (c) Arrow (->) (d) Indirection
8. Which of the following operator is not overloaded?
   (a) + (b) = (c):: (d) ,(Comma)
9. Templates enables us to create a range of related
   (a) classes (b) variables (c) Functions (d) Both (a) and (c)
10. In public derivation accessibility of members of base class undergo the following modifications in the derived class:
    (a) private becomes private (b) protected becomes private
    (c) protected becomes protected (d) None of these

Fill in the blanks:-

11. The read () and write function() functions handle data in _____ form.
12. The OMT symbol used for instance diagram is _____
13. _____ diagram shows the sequence of events as well as objects exchanging events.
14. The insulation of data from direct access by unauthorized functions is called _____.
15. A template function works with _____datatypes.

True or False:-

16. A function declared as static cannot access non -static class members.
17. It is an error to have a function with the same signature in both the super class and its subclass.
18. Objects are passed to a function by use of call by reference only.
19. A function template can have more than one template argument.
20. Aggregation is often called 'a kind of' relationship.

Section B

Attempt any Four and Each Question carry 5 Marks **(4x5=20)**

Q1 Write a C++ program of sorting using templates.
Q2. Explain this pointer with example.
Q3. Explain the concept of default arguments in C++.
Q4. Write a C++ Program which takes 10 numbers by the user(from Console) and from these numbers write even numbers in file EVEN.CPP and odd Numbers in file ODD.CPP
Q5. Explain different types of Aggregation.

Section C

Attempt any Four and Each Question carry 10 marks **(4x10=40)**

Q1. Design a class diagram for the following object classes, use association name where needed and you may add additional object classes.

School , Playground, Pricipal , School Board , classroom , book , student , teacher ,cafeteria , restroom, computer ,desk , chair , ruler , door , swing.

Q2. Design an Event Trace diagram of dynamic model for ATM.

Q3.What is functional modeling?Compare with Object and Dynamic Model.

Q4.Design a DFD for Billing System,

Q5. Write a short note on
(a) Specialization
(b) Generalization
(c) Link
(d) Association

GLA UNIVERSITY MATHURA

Course: B. Tech. 2nd yr Sem.-III
Subject: Environmental Studies
End Term Exam 2011-12
Paper Code: CEE-201 MM: 80 Time 2.5 h

Section-A

Q. 1 Attempt all questions (1 x 20)

(i) Elisa test is used to detect

(a) Cholera (b) Malaria (c) AIDS (d) Tuberculosis

(ii) The forest (conservation) Act extends to whole India exept

(a) Karnataka (b) Jammu and Kashmir

(c) Hariyana (d) Uttar Pradesh

(iii) The provision for environmental protection in the constitution were made under

(a) Article 48-A and Article 51-A (b) Article 27-B (h)
(c) Article 51-A and Article 58-A (g) (d) Article 21-A

(iv) The existence of ozone belt around the earth is mainly responsible for filtering out

(a) UV radiation (b) IR rays

(c) Cosmic rays (d) Microwaves

(v) Acid rain is composed of

(a) HNO_3, HCl , H_2SO_4 (b) H_3PO_4
(c) CH_3COOH (d) HCOOH

(vi) Study of trends in human population growth and predict of future growth is called -

(a) Biography (b) Kalography (c) Demography (d) Psychology

(vii) Most populated city in the world-

(a) Mexico (b) Tokyo
(c) Seuol (d) Mumbai

(viii) Solid waste management is best conducted by

(a) Inceration (b) Sanitary land fill
(c) Dumping into the sea (d) Vacuum press filter

(ix) The term ecosystem was proposed by-

(a) W.E. Odum (b) A.G. Tansley
(c) H.T. Odum (d) E.P. Odum

(x) The killer in Bhopal disaster was

(a) Carbaryl (b) Methyl isocyanate
(c) Aldrin (d) Accidental fire

(xi) Disease caused by eating fish contaminated with methyl mercury is

(a) Minamata (b) Hashimoto
(c) Bright disease (d) Osteoporesis

(xii) Profuse growth of aquatic vegetation that reduces the dissolved oxygen content is called

(a) Eutrophication (b) Algal boom
(c) Both (a) & (b) (d) Fermentation

(xiii) The maximum permissible noise level for residential areas near road traffic is

(a) 70 dB (b) 85 dB
(c) 100 dB (d) 120 db

(xiv) Environmental hazards associated with hydroelectric energy are

(a) Destruction of animal habitate (b) Submergence of vegetation
(c) Earth quake (d) All

(xv) Nitrogen transformation by bacteria involve-

(a) Nitrogen fixation (b) Nitrification
(c) Nitrate reduction and denitrification
(d) All process

(xvi) Main cause of deforestation

(a) Increasing human and livestock population (b) Agriculture
(c) Dam construction
(d) Mining

P.T.O.

(vii) Sustainable ecosystem consist

(a) Land , forest and water ecosystems which can be removed (b) Tropical forests
(c) Costal lakes
(d) Agriculture ecosystem

(xviii) Chipko movement is given by -

(a) Sunder lal Bahuguna (b) Chandi Prasad Bhatt
(c) Both (a) & (b) (d) Megha Patekar

(xix) Appiko movement related to the state

(a) Karnataka (b) Uttrakhand
(c) Uttar Pradesh (d) Madhya Pradesh

(xx) EIA is a

(a) Self corrective tool (b) Impact assessment tool
(c) Mitigative measure (d) All of the above

Q 2. Attempt any four questions (4 x 5)

(i) Differentiate the natural and anthropogenic type of environment.
(ii) Write some important functions of ecosystems .
(iii) What are the major issues related to fossil fuel.
(iv) Give brief note on hydroelectricity.
(v) Write some detrimental effects of transportation on environment.
(vi) Write short note on acid rain.

Q 3. Attempt any Three questions. Question (i) is compulsory carry 14 marks and remaining question of 13 marks. (14 + 2 x 13 = 40)

(i) What is pollution. Discuss the cause and effect of air pollution.
(ii) Discuss the major role of the government in the protection of environment.
(iii) What do you meant by ozone layer depletion and describe its mechanism.
(iv) Explain the ecological foot print, Bio capacity of earth and sustainable population.
(v) Discuss the Malthus theory of population growth. Explain the population explosion with diagram.